

# Passages - Helgoland

Attitude of openness: we should not seek to teach the world how it should be. Let's listen to the world instead, in order to learn from it how to think about it.

When Einstein objected to quantum mechanics by remarking that 'God does not play dice, Bohr responded by admonishing him to 'Stop telling God what to do? Which means: Nature is richer than our metaphysical prejudices. It has more imagination than we do.

An isolated object, taken in itself, independent of every interaction, has no particular state. At most we can attribute to it a kind of probabilistic disposition to manifest itself in one way or another. But even this is only an anticipation of future phenomena, a reflection of phenomena past, and only and always relative to another object.

The conclusion is revolutionary. It leaps beyond the idea that the world is made up of a substance that has attributes and forces us to think about everything in terms of relations.

This, I believe, is what we have discovered about the world with quanta.

The search for knowledge is not nourished by certainty: it is nourished by a radical absence of certainty.

Thanks to the acute awareness of our ignorance, we are open to doubt and can continue to learn and to learn better. This has always been the strength of scientific thinking - thinking born of curiosity, revolt, change. There is no cardinal or final fixed point, philosophical or methodological, with which to anchor the adventure of knowledge.

There are many different interpretations of Nagarjuna's text. The multiplicity of potential readings is testimony to its vitality and to the capacity of ancient texts to continue to speak to us. What interests us, anew, is not what the prior of a monastery in India was actually thinking nearly two thousand years ago. What interests us is the power of the ideas that emanate today from the lines he left; how these, enriched by generations of commentary, may open up new spaces for thought, intersecting with our culture and our knowledge.

This is the meaning of culture: an endless dialogue that enriches us by feeding on experiences, knowledge and, above all, exchanges.

Information is only counting the number of possible states of something.

Relative information is a measure of the physical correlation between two variables. Two variables have relative information if they can be in fewer states than the product of the number of states that each can be in.

All the information that we have about the world, considered externally, is in these correlations. Since all properties are relative properties, everything in the world does not exist

other than in this web of entanglement.

The fragmentation of points of view, the multiplicity of perspectives opened up by the fact that properties are only relative, is repaired, made coherent, by this consistency, which is an intrinsic part of the grammar of the theory. This consistency is the basis of the intersubjectivity that grounds the objectivity of our communal vision of the

world.

The implications for the relationship between what we see and the world, however, are remarkable. When we look around ourselves, we are not truly observing we are instead dreaming an image of the world based on what we know (including bias and misconception) and unconsciously scrutinizing the world to reveal any discrepancies which, if necessary, we will try to correct.

What we see is not a reproduction of the external world. It is what we expect, corrected by what we can grasp. The relevant input is not that which confirms what we already know, but that which contradicts our expectations.

A technical term for 'referring to something in our mental processes (promoted by the German philosopher and psychologist Franz Brentano) is 'intentionality: Intentionality is an important aspect of the notion of meaning and our whole mental life. There is a close relationship between what happens in thoughts and what happens 'outside of thoughts: what thoughts mean. There is a close relation between the word 'cat' and a cat; between a road sign and what the road sign signifies.

Two variables have 'relative information' if they can be in fewer states than the product of the number of states that each can be in.

'This notion of 'relative information' is purely physical. It is central in quantum physics: relative information is generated by the interactions that weave the world. Notice that it connects two different things, just as meaning does. But it isn't enough to understand meaning: the world

swarms with correlations, but the vast majority of these do not signify or mean anything. To understand meaning we need something else.

Mach does not see knowledge as a question of deducing or intuiting a hypothetical reality beyond sensations, but as the search for an efficient organization of our way of thinking about these sensations. The world that interests us, for Mach, is constituted by sensations. Any firm assumption about what lies 'behind' those sensations is suspect as being a form of 'metaphysics'.

Bogdanov replies to Lenin that his criticism misses the point. Mach's thought is not idealism, much less solipsism. The humanity that knows is not an isolated, transcendent subject; it is not the philosophical 'I' of idealism: it is real humanity, immersed in concrete history, part of the natural world. The 'sensations' are not 'within our minds' They are natural phenomena in the world: the form in which the world presents itself to the world. They do not come to a self that is separate from the world: they come to the skin, to the brain, to the neurons of the retina, to the receptors in our ears. These are elements of nature.